

1.write a short note on: Packet Sniffing.

Packet Sniffing

Definition:

Packet sniffing is a technique used to capture and analyze data packets traveling over a network.

Explanation:

- Data in a network is transmitted as packets
- A packet sniffer intercepts these packets
- It reads and analyzes the data

Types:

1. Passive sniffing
 - Monitors traffic without modifying it
 - Used in hub-based networks
2. Active sniffing
 - Interacts with the network to capture packets
 - Used in switch-based networks

Uses:

Legitimate uses:

- Network monitoring
- Troubleshooting
- Performance analysis

Malicious uses:

- Stealing passwords
- Capturing sensitive data
- Eavesdropping

One-line answer:

Packet sniffing is the process of capturing and analyzing network packets to monitor or intercept data.

Softwares

Packet Sniffing Tools:

- Wireshark: Traces malicious traffic back to the source
- Colasoft Capsa: Provides in-depth network analysis
- NetworkMiner: Passive tool that captures packets

- Kismet: Open-source tool for wireless network monitoring
- OmniPeek: Packet sniffer and network analyzer tool

Prevention:

- Use encryption (HTTPS, SSL/TLS)
- Use secure protocols
- Use network monitoring tools

